The Dual Reverse-Calculation Pattern: A Methodology for Auditing Compliance Narratives in Traditional Chinese Medicine Manufacturing

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Version 1.0 (2026-10-11). This version supersedes the one posted to SSRN on 2026-07-14. Since that posting the contribution has been repositioned from "an anomaly is documented" to "a mechanism already established in the accounting literature as *real earnings management* is measured in a setting where the output is a physical quantity and the specification is a published number"; the corresponding literature was added (Section 7.1, Section 10). No datum, computation, or detection rule changed. **Not peer reviewed.** A public comment window ran 2026-08-16 to 2026-09-30 (github.com/lizhuojunx86/huadian/discussions/19) and received 0 comments; the disposition log is in the repository. Version 1.0 marks the author's decision to freeze the text, not peer endorsement.

Abstract

Batch manufacturing records in Traditional Chinese Medicine (TCM) extraction often appear "perfect"—charging masses precise to 0.01 kg, constant extraction temperatures, fixed solvent-to-material ratios, and batch-to-batch yields that vary by hundredths of a percentage point. Such uniformity is physically implausible. The behavior behind it is not new: accounting research has documented for two decades that agents under external assessment pressure manage **real activities** rather than the books in order to just clear an observable threshold—*real earnings management* (Roychowdhury 2006; Cohen, Dey & Lys 2008; Zang 2012; reviewed in Pincus & Wellmeyer 2026). What is new here is the setting. In manufacturing, the managed quantity is a physical yield rather than a reported accrual, and the threshold is a published specification limit rather than an inferred analyst forecast—so the mechanism becomes measurable without having to be estimated from an industry-year model. This paper argues that TCM compliance narratives are products of **dual reverse calculation**, and that a single audit lens misses at least half of the back-calculation signal. We distinguish two co-existing but opposite mechanisms. The first is **culturally driven (F-002)**: because TCM quality standards set only lower bounds on content (no upper bound), the heuristic "higher content equals better quality" biases incoming-material assay figures *upward*. The second is **game-theoretic (F-004)**: front-line operators deliberately record adjustable indicators at the least conspicuous compliant position to preserve a safety margin against future high-baseline assessment and against quality-assurance (QA) investigation, biasing derived indicators such as yield *downward*. We propose an integrated detection framework—a three-layer structural model (F-001), sub-channel splitting (F-003), and a dual reverse-calculation decision tree—operationalized as candidate invariants using a sigma-floor and edge-band test. We stress two boundaries. The first is scope: the empirical material is a single enterprise and two products (n=2). Because the contribution claimed is a detection method for an already-established mechanism rather than the discovery of a new behavior, n=2 is the sample size of a method demonstration, not the evidentiary base of the mechanism—it limits how far the *calibration* travels, not whether the mechanism exists. The second is that **identification is not confirmation**—the method raises structured suspicion, it does not prove fraud. We connect the pattern to Goodhart's law, prospect theory, principal–agent theory, actor-network theory, and communities-of-practice scholarship.

Keywords: compliance narrative; reverse calculation; Goodhart's law; data integrity; Traditional Chinese

Medicine; GMP audit; agentic knowledge engineering; invariant detection; sigma-floor; game theory

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0. Thesis and Positioning

This is the first case-driven methodology paper in Layer 2 of the project's D-route. Where companion papers 00–07 abstract *engineering* methods, this paper abstracts a **domain identification method**. The intended readership is AI engineers working on compliance auditing, industry internal-audit and QA practitioners, and regulators and scholars.

What is and is not claimed as new. The underlying behavior is well established outside this domain. Since Roychowdhury (2006), the accounting literature has treated the manipulation of real operating activities to reach an observable target as a distinct and expected phenomenon, separate from manipulating the reported numbers; Graham, Harvey & Rajgopal (2005) found executives openly prefer the real action to the accounting one, and Zang (2012) showed the two are traded off against each other. Read through that literature, an operator who runs the process to land just above a specification floor is doing something a reader of that field would call unsurprising. **This paper does not claim to have discovered that behavior.** It claims three things that the accounting treatment cannot supply:

1. **A setting where the mechanism is directly observable rather than estimated.** Real-activities manipulation in financial data is inferred as a residual—abnormal cash flow, abnormal production cost, abnormal discretionary expense, each measured as a deviation from an industry-year model of what is "normal" (Roychowdhury 2006). In batch manufacturing the target is not inferred: the specification limit is a published number, the managed quantity is a physical yield recorded against it, and the deviation from the limit can be read off directly. **The layer that carries most of the contestability in the accounting literature — estimating what "normal" would have been — is not needed here.** A judgment call does remain, but it is a different and, we argue, a more tractable one: not *what level should this firm have reported*, but *how little batch-to-batch variance is physically possible for this process*. That question has a physical answer in principle, even though this paper answers it from an interview rather than from data (§7.1.1, §8).
2. **Directionality.** The accounting framing is one-directional—push the indicator over the threshold. This paper's central structural claim is that within one system two reverse calculations run in *opposite* directions (F-002 upward, F-004 downward) because they have different drivers, and that an audit lens tuned to one is blind to the other.
3. **An operational detection rule.** The framework is specified tightly enough to run: a sigma-floor and edge-band test, validated end-to-end on a real runtime (§9.1).

The honest form of the contribution is therefore **known mechanism × new setting × operational detection**, not a new finding about human behavior.

Core thesis:

> What AI knowledge engineering adds to compliance-narrative auditing is the ability to identify a **dual reverse-calculation mechanism** — not the observation that reverse calculation occurs, but the separation of it into two components: 1. **F-002, culturally driven** ("higher content = better quality" / unconscious / *upward* inflation); 2. **F-004, game-theoretically driven** (operator self-protection / conscious / *downward* edge-hugging).
>
> The two **co-exist but run in opposite directions, arise from different drivers, and require different identification methods**. A single identification frame—such as the traditional GMP

audit assumption that "the data are genuine, so anomalies should surface"—misses at least half of the reverse-calculation suspicion. Integrated identification requires the F-001 three-layer structure, F-003 sub-channel splitting, and a dual reverse-calculation decision tree.

Key differences from established GMP audit frameworks:

Dimension	PIC/S GMP	FDA Process Validation	This methodology
Data assumption	Genuine + traceable	Genuine + reproducibility verifiable	**Layered artifact** (F-001 / surface ≠ execution ≠ embodied experience)
Reverse-calc identification	Document review + deviation log	Control charts + Cpk	**Bidirectional morphology** (F-002 upward + F-004 downward)
Cross-batch $\sigma \approx 0$	Judged as excellent reproducibility	Very high Cpk = conforming	**Reverse-calculation smoking gun** (true σ has been flattened)
Interviews	Selective (e.g., during an FDA 483)	Selective	**First-class input / necessary condition for finalizing F-004**

A traditional framework that sees "six batches with yield $\sigma = 0.022$, hugging the 40% lower limit" would judge the process excellent; this methodology judges it a reverse-calculation suspect. This is a **fundamental inversion** of the judgment logic, and the basis for the inversion comes from interview ground truth, not from the data themselves.

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1. Motivation — Why Dual Reverse-Calculation Identification Is Needed

TCM batch records look "perfect" on their face: charging precise to 0.01 kg, constant extraction temperatures, a stable solvent-to-material ratio of 6.000, and yields across six batches confined to a 0.06 percentage-point band. Such perfection is almost physically impossible. The natural moisture content of decoction pieces fluctuates by ±2–5%; operator weighing error runs ±0.05 kg per bag; the electronic balance has ±0.5% precision; and the relative-density endpoint judgment in concentration has ±0.01 precision. Any one of these introduces deviations on the order of hundreds of grams. More suspicious still: for two different products (P1 and P2), with different processes and made in different periods, the charging masses are both "precise to 0.01 kg with cross-batch $\sigma = 0$." This can only be the product of reverse calculation back to a template batch size, not of genuine weighing. The case-2 data show that the records have undergone **bidirectional reverse-calculation processing**:

Dimension	Incoming-material inspection side	Batch-record side
Data morphology	Content hugs the LOQ (one marker glycoside 1.93×, another 1.89×)	Yield hugs the boundary / $\sigma \approx$ 0.02 pp
Direction	**Upward inflation**	**Downward edge-hugging**
Proposition	F-002 cultural discourse	F-004 game theory

The two directions are opposite, yet both are reverse calculation. F-002 ("high is good") cannot explain why yield would be pushed to the edge; F-004 ("hug the edge to pass") cannot explain why content would be inflated. A user interview (Q-027) directly reveals the F-004 mechanism:

> "Operators tend to bring the parameters that will be inspected into compliance, and then, even if the actual yield is higher, they will not write down the true figure—because they fear the company will later assess them against the higher yield… Front-line operators tend to hug the passing edge every time, so as to leave themselves enough room for future failure."

Q-028 further reveals the operating mechanism: **they do not directly alter the recorded value; instead they control the process so that the outcome lands in the target band.** (Output parameters are constrained by QC and downstream steps, so operators do not falsify them; input parameters have no independent verification, so they are back-filled.)

Contributions of this paper: (1) we argue that at least two independent reverse-calculation mechanisms exist within a compliance narrative, and that they run in opposite directions; (2) the two are identifiable, engineerable as detection logic, and transferable across domains; (3) we specify the dual-identification framework tightly enough to execute, and run it (§9.1); (4) we deepen the F-004 proposition—operators select the *least conspicuous compliant position*, not simply the lower limit. Note what is deliberately absent from this list: the existence of threshold-driven manipulation of real activities is not offered as a contribution, being established in the accounting literature (§0, §7.1.1).

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2. F-001 Three-Layer Structure — A Layered Model of the Compliance Narrative

```
Surface (compliance narrative): batch records + incoming inspection +
    release certificate → visible to GMP inspection
   ↓ structural generation gap
Middle (actual execution): much is never written into records / decided by
    the workshop head + scheduling / actual equipment availability
   ↓ structural generation gap
Deep (embodied experience): operators' tacit knowledge / self-protection
    strategy / true production noise
```

The surface layer comprises three parameter classes (triggered by Q-028; reverse-calculation cost increases across them):

Parameter class	Reverse-calculation mechanism	Example	External constraint
Input parameters	Back-filled into the record (no independent verification)	Charge 514.35 kg / temperature 100°C / solvent ratio 6.000	None
Output parameters	Genuinely recorded (not altered)	Relative density / assay / appearance	QC-independent + downstream
Derived indicators	Process-controlled to hug the edge (record not altered)	Yield ≈ 40% / 38.3%	Indirect (process control)

The middle layer is a level of operations-research decision-making (workshop head + scheduling; not personal preference, not explicit in the SOP). Two empirical forms appear: P1 exhibits **cross-batch switch-**

ing (T101 on equipment E1, 32 h → T102+ on equipment E2, 10 h) versus P2's **within-batch parallelism** (T301 runs three equipment sets in parallel, made explicit by SOP §4.10.5).

The deep layer contains true noise (loose execution) + inter-lot incoming variability + the self-protection strategy (the root of F-004) + shift coordination. The strict ABAB alternation of operators is reproduced 100% across 12 batches (P1 + P2)—the shift lead rotates assignments for wage fairness (Q-006). This is one of the few **genuine** fields in the surface layer.

The **three-layer generation gap** is the core of F-001: the surface $\sigma \approx 0$ (the narrative has been flattened by reverse calculation), the middle σ reflects true operations-research fluctuation, and the deep $\sigma_real > 0.5$ pp (true process noise). The larger the gap, the deeper the divergence between the compliance narrative and the actual execution. The first-principles task of AI knowledge engineering in compliance auditing is to **explicitly distinguish these three layers and measure the gap**—rather than feeding the surface record to a model as if it were ground truth. This is precisely where F-001 diverges fundamentally from the traditional "clean the data and you can model it" assumption (see companion paper 09).

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3. F-002 Cultural-Discourse Reverse Calculation — Upward Bias

Proposition: TCM quality standards set only a lower bound on content (no upper bound) → the "higher content = better quality" cultural discourse → an upward reverse-calculation bias. The **scope is restricted** to "content determination (one-sided lower bound)" for finished TCM and decoction pieces; it does not apply to intermediates, excipients, or packaging materials (which carry dual-limit standards, lack the cultural discourse; see ADR-034).

F-003 sub-channel splitting (finalized in three dimensions)—reverse-calculation difficulty $\propto 1 / |\text{standard value} \times \text{allowable error}|$:

Sub-channel	Standard magnitude	Measured multiple	Reverse-calc difficulty
main_component	≥ 30%	1.01–1.17× (hugging LOQ)	Low (an absolute difference of 1–4 pp already reads as "just over the line")
extract (leachables)	2–30%	1.25–1.68×	Medium
minor_-component	< 30%	1.30–2.83×	High (an absolute difference of 0.04–2 pp is, conversely, conspicuous)
dual_limit (intermediate/excipient/packaging)	Interval-based	Hugs the midpoint / consistent across lots	Low (F-002 does not apply)

Cross-product evidence: the most extreme main-component sample hugging the LOQ is P2's gypsum ($CaSO_4 \cdot 2H_2O$) at 1.014× (only 1.3 pp high); the minor sub-channel distributions across products **overlap almost completely** (P1 mean 1.839 vs P2 mean 1.833; difference 0.006; ranges overlap 78%). Main hugging and minor dispersed-but-co-distributed together constitute the cross-product confirmation of F-003.

The range-ratio matrix (the strongest evidence of sub-channel splitting): within a single lot (L1), the horizontal main:minor range ratio is approximately **55:1**; across lots (L1 → L2) it widens to **136–370:1** (minor range ~9.98 vs main range ~0.027). That is, the same reverse-calculation bias produces almost no visible fluctuation in the main-component channel (the absolute differences are small and easily "just over the line")

yet large dispersion in the minor channel (small absolute differences that are nonetheless conspicuous and hard to back-calculate). An F-001 second-order check that does not split by sub-channel will see the reverse-calculation signal diluted by cross-indicator averaging.

The strongest form of qualitative reverse calculation: gypsum's heavy-metal and arsenic results are reported only as "did not exceed X" with no numeric value ("did not exceed" is equivalent to "≤ upper limit," leaving no datum that can be independently scrutinized); and the mercury content across four materials (one material under P1; three under P2) is uniformly reported as 0.1 mg/kg (50% of LOD; A-018). Cross-species physiological bioaccumulation differences should span more than 100-fold, so a uniform fill value is an LOD-boundary reverse-calculation suspect.

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4. F-004 Game-Theoretic Reverse Calculation — Downward Edge-Hugging (Core of This Paper)

Proposition (emerged at GO-E-α; X1 finalized at GO-G-γ): out of game-theoretic self-protection, operators write adjustable indicators at the **least conspicuous compliant position**, leaving a "safety cushion" for future failure. The driver is twofold: fear of being assessed against a high baseline + fear of QA investigating an anomaly.

Deepening of the proposition—it is not a mechanical "hug the lower limit," but a choice of the "least conspicuous position":

Product	SOP spec	Hugged position	σ (pp)
P1	Interval 40–45% / no benchmark	Lower limit 40%	0.0223
P2	Interval 36–41% / benchmark 38.3%	Benchmark 38.3%	0.0275

Where the SOP gives an interval → operators choose the lower limit; where it gives a benchmark → they choose the benchmark value. Both forms are empirical evidence of "leaving room for failure." The σ magnitudes are consistent across products (ratio 1.23) and the deviation magnitudes are nearly identical (0.26–0.32 vs 0.264–0.341 pp), constituting a complete empirical demonstration of the cross-product robustness of F-004 X1.

F-002 vs F-004 comparison (core table):

Dimension	F-002	F-004
Reverse-calc direction	Upward inflation	Downward edge-hugging / benchmark-hugging
Applicability	Content lower-bound indicators (incoming)	Yield / derived indicators (batch records)
Driver	Cultural discourse ("high is good")	Game-theoretic self-protection (fear of assessment + fear of investigation)
Trigger	Mental bias (unconscious)	Rational strategy (conscious)
Data morphology	Multiple 1.04–2.83×	σ ≈ 0 + hugging the boundary
Identification method	Sub-channel distribution test	Cross-batch σ + boundary-hugging + **interview**
Operating mechanism	Directly fill high	Do not alter the record / process-control to target

Dimension	F-002	F-004
Agent	Inspector / process technician	Front-line operator / shift lead

Cross-product σ evidence (raw data for 12 batches / sample σ): P1, six batches, yields [40.27, 40.26, 40.30, 40.28, 40.32, 40.30] (σ = 0.0223, hugging the 40% lower limit); P2, six batches, yields [38.575, 38.564, 38.586, 38.581, 38.641, 38.575] (σ = 0.0275, hugging the 38.3% benchmark). The two products differ in process, equipment, and batch size, yet their σ values are nearly identical (magnitude ratio 1.23)—itself circumstantial evidence that σ is determined by recording/edge-hugging habit rather than by process fluctuation. Combined with cross-batch σ = 0 for the charging masses of seven materials, the solvent ratio, temperature, and time, this constitutes complete confirmation of the F-001 surface layer.

Key engineering implication: traditional statistical auditing looks only at distribution morphology ($\sigma \approx 0$ is treated as excellent); an F-004 determination must overlay interview ground truth (true process $\sigma > 0.5$ pp) to flip "excellent" into "reverse-calculated." This is the watershed between this methodology and traditional auditing, and it is the fundamental reason why, in the engineering of §5, the σ threshold must be drawn from interviews rather than set as a constant.

Cross-domain transfer value: F-004 is a proposition combining game theory and industrial psychology, applicable to any domain in which three conditions co-exist—external assessment + front-line self-entry authority + bounded indicators (earnings edge-hugging, p-hacking, KPI attainment, clinical endpoints). The proposition formalizes as: **the interaction of reverse-calculation difficulty + operator self-protection strategy + external assessment mechanism.**

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5. Integrated Framework — Dual Reverse-Calculation Identification

Identification decision tree:

```
flowchart TD
    A[Batch data + SOP interval] --> B{Indicator nature}
    B -->|Content lower bound| C[F-002 channel: sub-channel multiple +
        cross-lot σ]
    B -->|Process/yield/derived| F[F-004 channel: cross-batch σ +
        boundary-hugging]
    C --> C1{main σ << minor σ?}
    C1 -->|Yes| C2[F-003 sub-channel-splitting confirmed]
    F --> F1{σ ≈ 0 and hugging boundary?}
    F1 -->|Yes| F2[F-004 suspicion → interview confirmation needed]
    F1 -->|No| F3[actual]
    C2 --> Z{F-002 + F-004 dual?}
    F2 --> Z
    Z -->|F-002 only| Z2[Incoming reverse-calculated / process genuine]
    Z -->|F-004 only| Z3[Incoming genuine / process reverse-calculated]
    Z -->|Both| Z4[Strong compliance layer]
    Z -->|All actual| Z5[Lean toward ground truth]
```

This can be engineered into candidate invariants V12–V14 (extending the V1–V11 data-correctness invariants into the compliance-audit domain):

Invariant	Detection
V12 `reverse_calc_F002`	Test incoming lower-bound indicators by distribution across six sub-channels (single-lot multiple + cross-lot σ)
V13 `reverse_calc_F004`	Batch records with cross-batch σ ≈ 0 and deviation from the SOP boundary < 1 pp
V14 `dual_limit`	dual_limit sub-channel hugs the midpoint / consistent across lots

The key engineering contract: the σ threshold (a lower bound on true process noise) **comes from domain interview ground truth, not a constant**—this is the fundamental difference between this methodology and traditional statistical auditing. The V13 detection logic (deterministic, no LLM required):

```
def detect_f004(batches, spec, sigma_floor_pp):   # sigma_floor comes from
    interview (e.g., Q-027: >0.5pp)
    yields = [b.yield_pct for b in batches]
    sigma  = pstdev(yields)
    edge   = nearest_spec_edge(mean(yields), spec)         # lower limit /
        benchmark / midpoint
    if sigma < sigma_floor_pp / 20 and abs(mean(yields) - edge) < 1.0:
        return Suspicion("F-004", form=classify_edge(edge),
            needs_interview=True)
    return OK()
```

F-002 and F-004 are **two independent paths**; their conclusions feed into the decision tree for integrated judgment, and neither single path is complete on its own. The "suspicion" produced by detection enters human triage (interview / on-site confirmation), and the review conclusion is written back as the next round's ground truth—forming a closed loop of "structural check raises suspicion → interview-confirmation mechanism → ground truth written back."

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6. Applied Example — TCM Extraction Case (case-2)

Two products (P1 and P2) / 12 batches + 29 materials in 6 classes / 6 suppliers. Key findings:

- **F-002 (incoming side):** main hugs the LOQ (gypsum 1.014×, poria 1.07×); minor distributions overlap across products [1.30–2.83×]; qualitative suspicion (gypsum "did not exceed" + uniform mercury 0.1).
- **F-004 (batch-record side):** precise per-batch σ computation—

Product	Mean	Sample σ (pp)	Hugs	Flattening relative to true σ > 0.5 pp
P1 (6 batches)	40.288	0.0223	Lower limit 40.0	~24.6×
P2 (6 batches)	38.587	0.0275	Benchmark 38.3	~19.9×

- **F-001 three layers:** surface charging/yield/process parameters σ ≈ 0; middle layer two forms; deep layer ABAB stable across products + revealed by Q-027.

Integrated judgment = mixed compliance layer: incoming main + yield + process parameters = strong reverse-calculation suspicion; incoming minor + output parameters (QC-reviewed) = lean genuine; operators/shifts = genuine. "Mixed" is not vagueness but a **field-by-field layered** conclusion—within one and the same batch record, fields constrained by QC/downstream lean genuine while fields without external verification lean reverse-calculated. The implication for AI knowledge engineering is: **one cannot assign a single trustworthiness score to a whole record; trust must be assigned channel by channel (parameter class × sub-channel × presence of external constraint).** A traditional Cpk lens would judge a yield $\sigma = 0.022$ a "perfect process"; this methodology judges it a reverse-calculation smoking gun—a **judgment-logic inversion** whose basis comes from the Q-027 interview, not from the data themselves.

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7. Cross-Domain Transferability

Applicability conditions (all three co-existing): external assessment pressure + front-line self-entry authority + indicators with SOP boundaries.

§7.1 Earnings benchmark-beating — the strongest cross-domain isomorph of F-004. Burgstahler & Dichev (1997) found a discontinuity in the earnings of U.S. listed firms around the "zero" threshold: the frequency of small losses is anomalously low while small profits just above zero are anomalously high—pushing the indicator just over the boundary. Graham, Harvey & Rajgopal (2005), surveying 401 executives, found that roughly 78% admitted to sacrificing long-term value to meet targets, and that they **prefer real operating actions over accounting adjustments**—isomorphic to F-004's "change the process, not the books" (§4 / Q-028).

§7.1.1 Real earnings management is the field-standard name for F-004's mechanism. The preference Graham et al. documented in survey form was subsequently formalized as a research program in its own right. Roychowdhury (2006) operationalized *real activities manipulation* through three proxies—abnormal cash flow from operations, abnormal production costs (cost of goods sold plus the change in inventory), and abnormal discretionary expenses (advertising plus R&D plus SG&A)—each computed as the residual from an industry-year model of normal levels. Cohen, Dey & Lys (2008) showed that accrual-based management declined after the Sarbanes-Oxley Act while real-activities management rose, i.e. the pressure did not disappear, it relocated to where detection was harder. Zang (2012) established that managers trade the two off deliberately, choosing the cheaper instrument given their constraints. Cohen & Zarowin (2010) found real-activities manipulation around seasoned equity offerings has more severe consequences for subsequent performance than the accrual-based kind, and Gunny (2010) found firms that just meet benchmarks via real manipulation underperform afterward. Pincus & Wellmeyer (2026) review the post-SOX state of this literature, including evidence that external auditors constrain earnings management in settings of heightened pressure. On reading a draft of this paper's framing, Pincus characterized the behavior documented here as real earnings management and observed that it is "not a surprising result" for entities managing normal activities to just clear an externally observable target (M. Pincus, personal communication, August 18–19, 2026, cited with permission).

§7.1.2 What the accounting review says about auditors, and why it matters here. Two qualifications in Pincus & Wellmeyer (2026) bear directly on whether this methodology has an audience, and both cut against an easy reading:

1. **"Auditing standards do not place the detection of REM directly within the auditor's purview"** (Pincus & Wellmeyer 2026). Auditors are documented as *sensitive* to client REM—treating it as a negative signal about management attitudes, and letting it shape communications with audit committees

and client-retention decisions—but detecting it is not a standards-mandated duty. The implication for any detection method is unpleasant and worth stating plainly: **sensitivity without mandate does not reliably create demand.**

2. **"Unlike internal audits, external audits do not focus on operating efficiency or effectiveness"** (Pincus & Wellmeyer 2026). The financial auditor is institutionally pointed at the report, not at the process—which is precisely where real-activities manipulation lives. The same authors close the point by asking whether "closer coordination between external and internal auditors might enhance the ability to identify REM more reliably," and mark it as an open question for future research.

The GMP setting is an instance of the arrangement that open question asks about. A GMP quality audit is external to the audited party yet takes operations as its subject matter; data integrity of manufacturing records is inside its scope rather than adjacent to it. The separation that constrains REM detection in financial auditing—assurance function pointed at reports, operations expertise held internally—is not present in the same form. We do not claim this resolves their question, which is posed about financial auditing and would have to be answered there. We note only that a setting where the two functions are already combined exists, is regulated, generates records, and has so far been studied for compliance rather than for manipulation.

A third qualification from the same review cuts the other way, in this paper's favor. Citing Dichev et al. (2013), the authors note it can be difficult even for financial analysts to distinguish spending cuts made for business reasons from cuts made to hit a target. **That discrimination problem is much weaker here.** The threshold is not inferred from analyst forecasts or industry-year models; it is a numeric limit published in the pharmacopoeia. A yield sitting 0.1 pp above a published floor across six consecutive batches has few business-driven readings available to it.

Two consequences for this paper. First, terminological: **F-004 is a real-activities manipulation instance, and should be read as such.** A reader from accounting will find nothing surprising in the behavior; what this paper offers them is a setting, not a discovery.

Second, and more useful: **the two fields have opposite gaps, and they are complementary.** Accounting has roughly twenty years of validated estimators for real-activities manipulation but must infer the manipulation as a residual from a model of "normal," which is where the literature's methodological disputes concentrate. GMP manufacturing has the opposite profile—the target is published, the managed quantity is physically recorded, and no normality model is needed—but has **no estimator at all**; the incumbent method is document review by eye for handwriting and alteration. This paper is best understood as an attempt to carry the first field's question into the second field's data, where the measurement is cleaner. That transfer, rather than the behavioral claim, is where its risk and its value both sit.

A limit on the transfer. The accounting estimators rest on large cross-sections (thousands of firm-years) precisely because the residual is noisy. The corresponding leverage here is not sample size but directness. This is a real advantage for detection and a real disadvantage for calibration: a σ-floor of 0.5 pp is not derivable from theory and, in this paper, comes from an interview rather than a distribution (§5). Establishing σ-floors the way accounting established discretionary-accrual norms—empirically, across many producers—remains undone, and is the single most valuable thing a larger dataset would buy.

§7.2 Academic p-hacking — isomorph plus methodological caution. Simmons, Nelson & Simonsohn (2011) showed that undisclosed "researcher degrees of freedom" can push the false-positive rate to roughly 60%, by a mechanism isomorphic to F-004 (controlling process choices to cross the $p < 0.05$ threshold). A caution is warranted, however: it was once argued that "an anomalous clustering of p-values just below 0.05" is distribution-level evidence of p-hacking (Masicampo & Lalande 2012; Head et al. 2015), but subsequent reanalyses **failed to robustly replicate** this and noted that publication bias is a more likely explanation. This conversely confirms the core claim of this paper: **pure distributional evidence ($\sigma \approx 0$ / a bump at**

a threshold) is itself ambiguous; the reason case-2's F-004 is stronger is that it overlays the Q-027 interview as ground truth—**the distribution raises suspicion, the interview confirms the mechanism.** The cross-domain isomorphism of F-002 (cultural-discourse upward) is weaker and requires domain-by-domain identification of the discourse structure.

§7.3 Relationship to existing theory:

- **Goodhart's Law / Campbell's Law** (a measure that becomes a target ceases to be a good measure): F-002 + F-004 are a **structural extension** of these—Goodhart's original proposition argues only that "reverse calculation will occur," whereas this paper argues for its **directionality** (within a single system, upward and downward can co-exist, the direction determined by "indicator nature + driving mechanism").
- **Prospect Theory** (Kahneman & Tversky): the cognitive basis of F-004 is loss aversion—"reporting a high yield → being assessed against the high figure in future" is a potentially large loss, while "hugging the passing edge" is the rational, loss-avoiding choice even at the cost of forgoing short-term gain. F-004 further extends single-decision loss aversion into a stable behavioral pattern under **repeated-game** conditions (cross-batch $\sigma \approx 0$).
- **Principal–Agent** (information asymmetry): the F-001 three layers are a multi-level instantiation of it—the principal (company / regulator) sees only the surface inscription, while the agent (operator / process technician) alone knows the middle and deep layers; this is a structurally irreducible asymmetry. The policy implication: intensifying surface inspection is ineffective (the principal cannot see the middle and deep layers); one must change the information structure (a workshop observer / changing the assessment mechanism).
- **Latour ANT / STS:** the surface record is an **inscription** that engraves fluid practice into a circulable document; what GMP audit sees is the inscription black box. $\sigma \approx 0$ hugging the boundary is a **steady-state trace** of a trial of strength reached among the parties, not process excellence. This methodology can be read as an AI-engineering version of ANT—identifying the traces of strategy left on the inscription.
- **Wenger communities of practice / Scott métis:** F-004 is not an individual moral problem but a stable strategy transmitted via apprenticeship within a community of practice (the 100% cross-product reproduction of ABAB shift rotation is its organizational trace). Hence training / SOPs / warnings cannot eliminate F-004—it does not reside at the individual level; the path to its elimination lies in changing the middle-layer operations mechanism + the organizational-cultural structure (e.g., team performance replacing per-batch wages).

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8. Limitations

- **Inference boundary (n=2):** all evidence rests on a single enterprise and two products (P1 + P2 / 12 batches + 29 materials), and the interviews are indirect evidence (not on-site observation, not the operators themselves). The cross-product robustness conclusions are therefore limited to "internal consistency across the two products"; extrapolation across enterprises and across product categories is a **directional hypothesis, not a proven proposition** (X2, see the next item). **What this boundary does and does not touch (see §0, §7.1.1):** because the mechanism itself is established elsewhere—real-activities manipulation under threshold pressure is a documented finding in accounting, not a proposition this paper must carry—n=2 does not put the mechanism at risk. It puts the **calibration** at risk. The σ-floor and edge-band constants are fitted to one producer's practice; whether 0.5 pp is a general threshold or an artifact of this workshop is exactly what n=2 cannot answer, and it is the first

thing a second enterprise would settle.

- The F-004 X2 cross-enterprise empirical test has not been performed (a decision was made to set it aside for now / documented as unresolvable / generalizability to be strengthened via literature review).
- We lack "second-order ground truth" such as raw HPLC chromatograms or TLC plate photographs.
- **Identification ≠ confirmation (a methodologically self-aware boundary):** dual reverse-calculation identification is a **structural screen that raises suspicion, not a conviction.** $\sigma \approx 0$, boundary-hugging, sub-channel splitting, and the like only flag a field as a "reverse-calculation suspect"; confirmation must overlay interviews / on-site observation / raw instrument data (the closed loop of §5, and the lesson of §7.2 that "distributional evidence is itself ambiguous"). This methodology is positioned as the **first screen** of an audit; it does not replace institutional confirmation.
- Ethics: F-004 is the **rational behavior** of front-line personnel, not a moral problem; technical means (AI auditing) cannot substitute for institutional design. The cross-cultural universality of F-002 is lower than that of F-004.
- **No national specificity is claimed, and the reader should not infer one.** F-004 is an instance of a mechanism documented for two decades in the accounting literature, principally on U.S. listed firms (Burgstahler & Dichev 1997; Graham, Harvey & Rajgopal 2005; Roychowdhury 2006; Cohen, Dey & Lys 2008), where the same behavior—managing real activities to just clear an externally observable threshold—is treated as expected rather than aberrant. Nothing in this paper suggests that the operators or the enterprise studied here are less honest than counterparts elsewhere, and the framework would predict the same behavior wherever its three applicability conditions hold (§7). The TCM extraction setting is used because the **measurement** is unusually clean—the specification limit is published and the managed quantity is physically recorded—not because the **conduct** is unusual. F-002 is the one component with a genuinely domain-specific origin, and its scope is stated accordingly: it follows from the lower-bound-only structure of the relevant content standards, not from any cultural claim about practitioners.

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9. Relationship to Existing Methodology

§9.1 Implementing V12–V14 in traceguard (the runtime substrate)

> **[PoC-validated 2026-06-04 / GO-Z-η]** The E1–E3 + V13 design in this section has been landed on the real traceguard architecture and run end-to-end (a real 6-batch σ matrix run through the real CLI `guardian check` → `action=alert` / $\sigma = 0.0203$ hugging the 40.0 lower limit / DEGRADED with no LLM / dogfood 246 passed / 0 regressions). The real run corrected naming and units relative to the design draft: the method name is `output.output_as_dict()` (not `as_dict()`); core mode uses the generic `"sigma_floor"` (dropping the `F004_` prefix, with F-004 semantics moved into a config comment); and core fields use `sigma_floor` / `edge_band` (dropping the `_pp` unit, with pp moved into a config comment). **A new E4 candidate = structured-output persistence:** `EvalTrace` stores only a truncated `output_preview`, so cross-batch scalar retrieval requires structured persistence—something a pure design draft could not have surfaced; it has been registered as a candidate input for the D-route 2027-01 framework v0.1. See `../cases/tcm-extraction/traceguard-poc/poc-results-v0.1.md`.

The two Layer-1 substrates divide labor: the HuaDian knowledge-side framework (`identity_resolver` / `invariant_scaffold` / `audit_triage`) and the traceguard runtime QA substrate are complementary. V12–V14 are deterministic statistical checks that land in traceguard's **structural layer** (`validators/structural.py::validate_structural → StructuralResult`), **requiring no**

LLM—they run as usual in `env.py`'s DEGRADED mode (structural-only). Mapping points:

- Cross-batch σ needs historical data → read the eval_store via `store/reader.py::TraceReader.query_traces()` (state lives in the eval_store; guardian remains stateless);
- Suspicion is persisted → `TraceWriter.write(action="alert", passed=False, issues=[...])` (reverse calculation is an audit flag, not an abort);
- Review → `optimizer/suggestion.py::generate_suggestions()` (advisory / human-in-the-loop / never auto-applied);
- σ_floor / sub-channel definitions / spec boundaries → domain config (`configs/examples/tcm_extraction.yaml`), not the generic core.

The stateless × cross-batch-σ solution: the traceguard guardian is per-step stateless, but V13 needs to span six batches. Following the principle "all state lives in eval_store," V13 pulls the prior N batches via `TraceReader.query_traces(limit=6)`, computes σ in memory, and writes back—so guardian itself remains stateless. Domain parameters (including the interview-derived σ_floor) go in config, not the generic core:

```yaml
# configs/examples/tcm_extraction.yaml (reverse_calc extension block /
    PoC-validated as-built naming)
reverse_calc:
  mode: "sigma_floor"          # generic / core does not contain F004; the
      F-004 downward edge-hugging semantics are noted here
  window_batches: 6
  sigma_floor: 0.5             # pp / (*) ground truth from interview
      Q-027 (not a constant / core field drops the pp unit)
  spec_edges: { "P1": {type: interval_low, value: 40.0},
                "P2": {type: benchmark, value: 38.3} }
```

The implementation requires three framework extension points (= case-2's framework-abstraction output for traceguard / D-route 2027-01 v0.1 candidates): **E1** a new `reverse_calc` structural-check type (a generic statistical reverse-calculation capability, not bound to TCM); **E2** pluggable check registration (currently four are hard-wired / they need to be enabled by config declaration); **E3** "suspicion vs failure" audit-flag semantics (`passed=False` but the data are not "wrong," and must not pollute `get_step_stats().pass_rate`). These three are a textbook separation of generic capability into core and domain parameters into config. The full implementation design (mapping table + V13 real-signature pseudocode + closed loop) is in the unabridged draft, archive §9.1.

§9.2 Relationship to existing framework papers: V12–V14 extend paper 04 (invariants); F-002/F-004 suspicion → the triage of paper 05 (audit trail); cross-lot/batch linkage uses R1–R6 from paper 03 (identity resolver); ADR-033/034/035 are instances of paper 06 (ADR for KE). **§9.3 Relationship to the Layer-3 case library:** this paper is the methodological distillation of case-2, and other cases (earnings, clinical, etc.) are encouraged to cite, validate, and refute it. With paper 09 (layered compliance narrative / structural identification), it forms a "structure × mechanism" mutual-confirmation panel.

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10. References

All entries have been verified by web search; volume, issue, and page numbers have been finalized. Goodhart's (1975) original (RBA *Papers in Monetary Economics*) has no standard pagination, so the precise volume and pages of the authoritative reprint are appended. The real-earnings-management entries added on 2026-08-19 were verified against publisher-deposited

Crossref records, cross-checked against RePEc and, for Roychowdhury (2006), against the article itself; the one field that could not be corroborated (final volume/issue/pages for Pincus & Wellmeyer, not yet assigned) is flagged in place rather than filled.

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11. Notes on Data and Reproducibility

This paper and all case-2 accompanying data have undergone systematic de-identification. Product names (P1/P2/P3), batch numbers (B1–B12), incoming-material lots (L1/L2), equipment models (E1–E6), and enterprise, supplier, and personnel names are all pseudonyms. Computation-input values—σ, range ratios, multiples, and the like—retain their original values, so the conclusions are recomputable. The pseudonym mapping is not published.

The source corpus is a TCM extraction case (a single enterprise, two products P1 + P2; 28 incoming-material inspection records + 12 batch manufacturing records + a 5-question user interview). Associated findings: F-001 / F-002 / F-003 / F-004. Associated decision records: ADR-033 / ADR-034 / ADR-035. The companion paper is "The Layered Compliance-Narrative Pattern" (paper 09), which provides "structure × mechanism" mutual confirmation.

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Data availability: The de-identified structured dataset underlying this paper—including the per-batch yield series and σ computations, the sub-channel multiple distributions, and the range-ratio matrix—is openly available under CC BY 4.0 in the project repository (directory `docs/cases/tcm-extraction/`): https://github.com/lizhuojunx86/huadian. The pseudonym mapping that links published values to the enterprise, products, batches, suppliers, or personnel is not published and will not be shared under any terms. Further de-identified numeric detail is available from the author on reasonable request, subject to the same de-identification constraints.

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